Beginner FinTech AI/ML Intern Training Syllabus

Duration: 3 Months | Structure: 1 Month Python → 1 Month FastAPI → 1 Month AI/ML

Month 1: Python Fundamentals

Week 1: Python Fundamentals

- Variables and type conversion
- Data structures (lists, tuples, sets, dictionaries)
- Comprehensions (list, dictionary, set comprehensions)
- Control Flow (conditional statements, loops: for/while, break, continue, pass)
- **Project:** Command-line Expense Tracker with categorization, budget management, transaction history (add/update/delete), total spending, category-wise summary, and alerts for transactions exceeding budget

Week 2: Object-Oriented Programming

- Classes, objects, constructors, instance vs. class variables
- Methods (instance, class, static)
- OOP fundamentals (inheritance, method overriding, polymorphism, encapsulation, properties, decorators)
- Common dunder methods
- Exception handling and file handling
- **Project:** Digital Wallet Transaction System with user management/tiers, multiple transaction types (deposit, withdrawal, transfer, payment), transaction validation, balance management, file-based storage, and exception handling

Week 3: Exploratory Data Analysis

- **Pandas:** DataFrame basics, selecting/filtering data, handling missing data, cleaning, grouping, aggregation, and data transformation
- **NumPy:** Mathematical functions, axis operations, aggregations, array concepts
- **Matplotlib:** Basic plotting workflow, visualization techniques, and parameters
- **Project:** FinTech Transaction Data Analysis in Jupyter Notebook covering missing data cleanup, transaction amount/frequency analysis, high-value transaction identification, spending patterns, customer/category aggregation, and trend visualizations

Week 4: Context Managers and Database

- Context managers (`with` statement, built-in managers, custom managers with `__enter__`/`__exit__`, `contextlib`)
- Relational database concepts and SQL fundamentals
- **SQLAlchemy:** Database models, connections, CRUD operations, and relationships (one-to-one, one-to-many, many-to-many)
- **Project:** Financial Transaction Database system using SQLAlchemy and context managers for session management, managing customers, accounts, transactions, categories, statuses, balances, and complex queries

Month 2: FastAPI

Week 5: REST API and FastAPI Fundamentals

- REST API concepts, HTTP request/response cycle, methods (GET, POST, PUT, PATCH, DELETE), and status codes
- FastAPI application structure, routes, endpoints, path/query parameters, request body, responses, and Swagger/OpenAPI documentation
- **Project:** Digital Wallet REST API converting the Week 2 system into endpoints for user/account creation, deposit, withdrawal, transfer, payment, transaction history, balance, and filtering with proper status codes

Week 6: API and Backend Architecture

- FastAPI dependency injection, standard project structure, configuration, and environment variables
- Basic asynchronous programming (`async` and `await`)
- Design patterns: Creational (Singleton, Factory, Builder), Structural (Adapter, Decorator, Facade), Behavioral (Observer, Strategy, Command)
- **Project:** Payment Processing Backend service supporting multiple payment methods with async operations, employing Factory (processor creation), Strategy (payment methods), Facade (workflow), and Observer (status notifications) patterns

Week 7: Authentication and Production Practices

- Authentication vs. authorization, password hashing, PASETO
- Middleware, background tasks, and API testing with `pytest`
- Basic API security and CORS
- **Project:** Secure Banking API applying PASETO authentication and authorization to secure the API endpoints built in previous weeks

Week 8: AI and Machine Learning Fundamentals

- Types of Machine Learning (supervised, unsupervised, reinforcement learning)
- Machine Learning workflow (dataset, features, labels, training, validation, testing)
- **Project:** Financial Transaction Risk Classification defining a risk problem, preparing features/labels, training a basic classification model (normal vs. risky), and evaluating training vs. testing results

Month 3: AI/ML Fundamentals

Week 9: Data Preparation and Supervised Learning

- Train/test split, feature selection, feature scaling, encoding categorical data
- Regression (Linear Regression) and Classification (Logistic Regression, Decision Tree, Random Forest)
- Scikit-learn model training, prediction, and pipelines
- Model evaluation metrics (Accuracy, Precision, Recall, F1 Score, Confusion Matrix, MAE, MSE, RMSE)
- Model problems: overfitting, underfitting, bias vs. variance
- **Project:** Credit Risk Prediction model predicting high-risk vs. low-risk borrowers using preprocessing, feature selection/scaling, multiple models comparison, metric evaluation, and overfitting analysis

Week 10: Deep Learning Fundamentals

- Neural network basics with TensorFlow/Keras, building and training simple neural networks
- Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs)
- **Project:** User spending prediction model using time-series forecasting with RNNs

Week 11: Introduction to Generative AI

- Large Language Models (LLMs), tokens, prompt engineering basics
- Introduction to embeddings and Retrieval-Augmented Generation (RAG)
- **Project:** FinTech Financial Assistant using LLM integration, prompt engineering, financial document embeddings, vector-based document retrieval, and a RAG pipeline for financial QA

Week 12: Final Project

- **Project:** AI-Powered Financial Risk & Transaction Analysis System — An end-to-end FinTech AI/ML application combining:
  - **Data Layer:** Transaction data, customer/account info, historical records
  - **ML Layer:** Preprocessing, feature engineering, risk classification, spending prediction, model evaluation
  - **AI Layer:** LLM integration, financial document understanding, embeddings, RAG-based QA
  - **Backend Layer:** FastAPI with PASETO authentication, DB integration, ML prediction & AI assistant endpoints
  - **Deliverables:** Working FastAPI backend, database, trained ML models, APIs, GenAI/RAG module, Swagger docs, unit/API tests (pytest), clean Git repository, and documentation